# Problem Set 5

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```r
seed_string <- "Roshie, Angie, Blair, Thomas, Aaron, John"
seed_number <-
  seed_string |>
  rlang::hash() |> # hash the string to a hex number
  substr(1, 7) |> # make the number smaller
  strtoi(base = 16) # convert to decimal

set.seed(seed_number)
```

## Hamilton's Problems

### Exercise 3.1

Is the following MA(2) process covariance-stationary? If so, calculate its autocovariances.

$$Y_t = (1 + 2.4L + 0.8L^2)\varepsilon_t$$

$$\mathbb{E}[\varepsilon_t \varepsilon_\tau] = \begin{cases} 1 & \text{for } t = \tau \\ 0 & \text{otherwise} \end{cases}$$

**Solution:** We check the conditions for stationarity. First, we take $\varepsilon_t \sim WN(0, 1)$.

$$\begin{aligned} \mathbb{E}[Y_t] &= \mathbb{E}[\varepsilon_t + 2.4\varepsilon_{t-1} + 0.8\varepsilon_{t-2}] \\ &= \mathbb{E}[\varepsilon_t] + 2.4\mathbb{E}[\varepsilon_{t-1}] + 0.8\mathbb{E}[\varepsilon_{t-2}] \\ &= 0 \quad \forall t \end{aligned}$$

by virtue of $\varepsilon_t$ being a white noise distribution centered at 0. Next, we check that the variance of this process is finite for all $t$.

$$\begin{aligned} \mathbb{V}(Y_t) &= \mathbb{V}(\varepsilon_t + 2.4\varepsilon_{t-1} + 0.8\varepsilon_{t-2}) \\ &= \mathbb{V}(\varepsilon_t) + 2.4^2\mathbb{V}(\varepsilon_{t-1}) + 0.8^2\mathbb{V}(\varepsilon_{t-2}) \\ &= 1 + 2.4^2 + 0.8^2 = 7.4 < \infty \quad \forall t, \end{aligned}$$

where we use the fact that the covariance of $\varepsilon_t$ and any $\varepsilon_\tau$, $\tau \neq t$ is 0, and that $\varepsilon_t$ has variance of 1 (for simplicity, it could be any finite $\sigma^2$).

Finally, we check that the covariance between $Y_t$ and some arbitrary $Y_{t-j}$ is dependent only on the size of the gap $j$.

$$\begin{aligned} \mathbb{C}\text{ov}(Y_t, Y_{t-j}) &= \mathbb{E}[Y_t Y_{t-j}] - \mathbb{E}[Y_t]\mathbb{E}[Y_{t-j}] \\ &= \mathbb{E}[Y_t Y_{t-j}] - 0. \end{aligned}$$

- $j = 0$: $\mathbb{C}\text{ov}(Y_t, Y_t) = \mathbb{V}(Y_t) = 7.4$

*Authors listed alphabetically by and contributed equally to every question.

- $j = 1$: $\mathbb{E}[Y_t Y_{t-1}] = \mathbb{E}[(\varepsilon_t + 2.4\varepsilon_{t-1} + 0.8\varepsilon_{t-2})(\varepsilon_{t-1} + 2.4\varepsilon_{t-2} + 0.8\varepsilon_{t-3})]$. Using linearity of expectations and the fact that $\mathbb{E}[\varepsilon_t \varepsilon_\tau] = 0$ for $t \neq \tau$, this simplifies to $2.4\mathbb{E}[\varepsilon_{t-1}^2] + 2.4(0.8)\mathbb{E}[\varepsilon_{t-2}^2] = 4.32$, independent of $t$.
- $j = 2$: $\mathbb{E}[Y_t Y_{t-2}] = \mathbb{E}[(\varepsilon_t + 2.4\varepsilon_{t-1} + 0.8\varepsilon_{t-2})(\varepsilon_{t-2} + 2.4\varepsilon_{t-3} + 0.8\varepsilon_{t-4})]$. Again, we use linearity of expectations and see that the only expectation that isn't 0 is the term $\mathbb{E}[\varepsilon_{t-2}^2]$; $\mathbb{E}[Y_t Y_{t-2}] = 0.8\mathbb{E}[\varepsilon_{t-2}^2] = 0.8$, independent of $t$.
- $j > 2$: $\mathbb{C}\text{ov}(Y_t, Y_{t-j}) = 0$ in this case, as there will be no overlapping $\varepsilon$ terms in the expectation of the product calculation. There is only the situation where $\mathbb{E}[\varepsilon_t \varepsilon_\tau], t \neq \tau$. Thus, $\mathbb{C}\text{ov}(Y_t, Y_{t-j}) = \gamma_j$, independent of $t$. The MA(2) process $Y_t$ satisfies all three conditions for covariance-stationarity. As an aside, this process is not invertible: the roots of the lag polynomial $P(z) = 0.8z^2 + 2.4z + 1$ are $-0.5, -2.5$, which are not both outside of the unit circle.

The auto-covariances are as described above (again, normalizing $\varepsilon_t$ to have variance of 1:

$$\mathbb{C}\text{ov}(Y_t, Y_{t-j}) = \begin{cases} 7.4 & \text{if } j = 0 \\ 4.32 & \text{if } j = 1 \\ 0.8 & \text{if } j = 2 \\ 0 & \text{else.} \end{cases}$$

## Exercise 3.2

Is the following AR(2) process covariance-stationary? If so, calculate its autocovariances.

$$(1 - 1.1L + 0.18L^2)Y_t = \varepsilon_t$$
$$\mathbb{E}[\varepsilon_t \varepsilon_\tau] = \begin{cases} 1 & \text{for } t = \tau \\ 0 & \text{otherwise.} \end{cases}$$

**Solution:**

We can rewrite $Y_t$ in the form

$$\xi_t = \mathbf{F}\xi_{t-1} + \eta_t.$$

Then,

$$\begin{bmatrix} y_t \\ y_{t-1} \end{bmatrix} = \begin{bmatrix} 1.1 & -0.18 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ y_{t-2} \end{bmatrix} + \begin{bmatrix} \varepsilon_t \\ 0 \end{bmatrix}$$

We find the eigenvalues of $\mathbf{F}$ by solving

$$\begin{aligned} \det \begin{pmatrix} 1.1 - \lambda & -0.18 \\ 1 & -\lambda \end{pmatrix} &= 0 \\ -1.1\lambda + \lambda^2 + 0.18 &= 0 \\ \lambda^2 - 1.1\lambda + 0.18 &= 0 \\ (\lambda - 0.9)(\lambda - 0.2) &= 0. \end{aligned}$$

Therefore, the two eigenvalues are $\lambda_1 = 0.2$ and $\lambda_2 = 0.9$. Because

$$\max_i |\lambda_i| < 1$$

we know that this AR(2) process admits a covariance-stationary MA($\infty$) process.

To calculate the mean and autocovariance, we proceed with the eigendecomposition of $\mathbf{F}$, which is:

$$\mathbf{T\Lambda T}^{-1} = \begin{pmatrix} 0.2 & 0.9 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0.2 & 0 \\ 0 & 0.9 \end{pmatrix} \begin{pmatrix} -\frac{10}{7} & \frac{9}{7} \\ \frac{10}{7} & -\frac{2}{7} \end{pmatrix}$$

Then,

$$\begin{aligned} F_{1,1}^{j} &= 0.2^2\left(-\frac{10}{7}\right) + 0.9^2\left(\frac{10}{7}\right) \\ &= \left(\frac{-2}{7}\right)0.2^j + \left(\frac{9}{7}\right)0.9^j. \end{aligned}$$

If $\psi_j = \left(-\frac{2}{7}\right)0.2^j + \left(\frac{9}{7}\right)0.9^j$, then

$$Y_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}.$$

We can see that

$$\mathbb{E}[Y_t] = 0.$$

For the autocovariance,

$$\begin{aligned} \gamma_j &= \mathbb{E}[(Y_t - \mu)(Y_{t-k} - \mu) \\ &= \mathbb{E}(Y_t Y_{t-k}) = \mathbb{E}\left[\left(\sum_{j=0}^{\infty}\psi_j\varepsilon_{t-j}\right)\left(\sum_{l=0}^{\infty}\psi_l\varepsilon_{t-l}\right)\right] \\ &= \mathbb{E}\left[\sum_{n=0}^{\infty}\sum_{j=0}^{n}\left(\psi_j\varepsilon_{t-j}\right)\left(\psi_{n-j}\varepsilon_{t-(n-j)}\right)\right] \text{(by Cauchy product)} \\ &= \sum_{n=0}^{\infty}\sum_{j=0}^{n}\psi_j\psi_{n-j}\mathbb{E}[\varepsilon_{t-j}\varepsilon_{t-n+j}] \end{aligned}$$

We know that $\mathbb{E}[\varepsilon_{t-j}\varepsilon_{t-n+j}] = 1$ when the indexes are equal to each other $t - j = t - n + j$, which occurs when $n = 2j$. Suppose $n = 0$, then $j = 0$ and we have $\psi_0\psi_0$. Suppose $n = 2$, then $j = 1$ and we have $\psi_1\psi_1$, etc.

Thus, we can observe that the above expression simplifies to

$$\begin{aligned} \gamma_j &= \sum_{j=0}^{\infty}\psi_j^2\mathbf{1}\{j \text{ is even}\} \\ &= \sum_{j=0}^{\infty}\left[\left(-\frac{2}{7}\right)0.2^j + \left(\frac{9}{7}\right)0.9^j\right]^2\mathbf{1}\{j \text{ is even}\} \end{aligned}$$

## Exercise 3.3

A covariance-stationary $AR(p)$ process

$$(1 - \phi_1 L - \phi_2 L^2 - ... - \phi_p L^p)(Y_t - \mu) = \varepsilon_t$$

has an $MA(\infty)$ representation given by

$$\begin{aligned} (Y_t - \mu) &= \psi(L)\varepsilon_t, \text{ with} \\ \psi(L) &= 1/[1 - \phi_1 L - \phi_2 L^2 - ... - \phi_p L^p] \text{ or} \\ [1 - \phi_1 L - \phi_2 L^2 - ... - \phi_L^p][\psi_0 + \psi_1 L + \psi_2 L^2 + ...] &= 1. \end{aligned}$$

In order for this equation to be true, the implied coefficient on $L^0$ must be unit and the coefficients on $L^1, L^2, ...$ must be zero. Write out these conditions explicitly and show that they imply a recursive algorithm for generating the $MA(\infty)$ weights $\psi_0, \psi_1, ....$ Show that this recursion is algebraically equivalent to setting $\psi_j$ equal to the $(1,1)$ element of the matrix $\mathbf{F}$ raised to the $j$th power.

**Solution:**

If we expand the product and make explicit the conditions for the $MA(\infty)$ representation to hold true, we obtain a series of equations on the respective lag terms:

$$
\begin{aligned}
&L^0 : \psi_0 = 1 \\
&L^1 : \psi_1 - \psi_0\phi_1 = 0 \Rightarrow \psi_1 = \phi_1 \\
&L^2 : \psi_2 - \phi_1\psi_1 - \phi_2\psi_0 = 0 \Rightarrow \psi_2 = \phi_2 + \phi_1^2 \\
&... \\
&L^j : \psi_j = \phi_1\psi_{j-1} + \phi_2\psi_{j-2} + ... + \phi_p\psi_{j-p} \\
&\quad \psi_j = \sum_{k=1}^{p} \phi_k\psi_{j-k}.
\end{aligned}
$$

As in lecture, we can define the **F** matrix as

$$
\begin{bmatrix}
\phi_1 & \phi_2 & ... & \phi_{p-1} & \phi_p \\
1 & 0 & ... & 0 & 0 \\
0 & 1 & ... & 0 & 0 \\
\vdots & \vdots & \vdots & \vdots & \vdots \\
0 & 0 & ... & 1 & 0,
\end{bmatrix}
$$

where the first row is $\phi_1$ through $\phi_p$, the other elements of the final column are 0's, and the remainder of the matrix is $I_{p-1}$ (the identity).

Then, $F_{11} = \phi_1 = \psi_1$, by construction. The (1,1) element of $F^2$ is $\phi_1^2 + \phi_2 = \phi_1\psi_1 + \phi_2 = \psi_2$, as desired. The (1,1) element of $F^p$ is $\phi_1\psi_{p-1} + \phi_2\psi_{p-2} + ... + \phi_p\psi_0$, which from the $MA(\infty)$ conditions we have shown to be $\psi_p$. In particular, the first two elements of the first row of the product $F^{j-1}$ are constructed as follows: $[\psi_{j-1}, \phi_2\psi_{j-2} + \phi_3\psi_{j-3}, ...]$, which when taking the dot product with $F$ leaves $\sum_{k=1}^{p} \phi_k\psi_{j-k} = \psi_j$ as the (1,1) element. Thus, the recursion algorithm is equivalent to setting $\psi_j$ to be the (1,1) element of $F^j$.

## Exercise 11.5

Consider the following bivariate VAR:

$$
\begin{aligned}
y_{1,t} &= 0.3y_{1,t-1} + 0.8y_{2,t-1} + \varepsilon_{1,t} \\
y_{2,t} &= 0.9y_{1,t-1} + 0.4y_{2,t-1} + \varepsilon_{2,t}
\end{aligned}
$$

with $\mathbb{E}[\varepsilon_{1,t}\varepsilon_{1,\tau}] = 1$ for $t = \tau$ and 0 otherwise, $\mathbb{E}[\varepsilon_{2,t}\varepsilon_{2,\tau}] = 2$ for $t = \tau$ and 0 otherwise, and $\mathbb{E}[\varepsilon_{1,t}\varepsilon_{2,\tau}] = 0$ for all $t, \tau$.

(a) Is this system covariance-stationary?

(b) Calculate $\mathbf{\Psi}_s = \frac{\partial y_{t+s}}{\partial \varepsilon_t'}$ for $s = 0, 1, 2$. What is the limit as $s \to \infty$?

(c) Calculate the fraction of the MSe of the two-period-ahead forecast error for variable 1,

$$\mathbb{E}[y_{1,t+2} - \mathbb{E}[y_{1,t+2}|y_t, y_{t-1}, ...]]^2,$$

that is due to $\varepsilon_{1,t+1}$ and $\varepsilon_{1,t+2}$.

**Solution:**

(a) This system is not covariance-stationary. We can rewrite it in vector form:

$$
\begin{bmatrix} Y_{1,t} \\ Y_{2,t} \end{bmatrix} = \begin{bmatrix} 0.3 & 0.8 \\ 0.9 & 0.4 \end{bmatrix} \begin{bmatrix} Y_{1,t-1} \\ Y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{bmatrix}.
$$

The eigenvalues of the $\pi$ matrix (coefficient on $Y_{t-1}$) are found by setting the determinant of $\pi - \lambda I$ to be 0; we get $(0.3 - \lambda)(0.4 - \lambda) - 0.8 * 0.9$, yielding the eigenvalues of 1.2 and -0.5. One eigenvalue of $\pi$ (the companion form) is outisde of the unit circle, meaning that this bivariate VAR process is not covariance-stationary.

(b)

# The Permanent Income Model

## 1. Finding the Autocovariances

Conditional on the information available at time $t$, all of the future random variables are mean zero, so the agent's expected income in any period $j \geq 0$ will their current permanent component of income, $x_t$. The transitory shock includes an $\epsilon_t$ that moves current period income from its long run average. So $E_t[y_{t+j+1}] = x_t$. The time $t+1$ expectations are themselves a random variable from the perspective of period $t$ because they depend on the news revealed in period $t+1$. In particular, in period $t+1$, a new value $x_{t+1}$ will be revealed, recentering the agents expectations of all future periods. If $j = 0$, then from period $t+1$'s perspective, $y_{t+1}$ is observed with certainty. But all future $j \geq 1$ are centered at $x_{t+1}$.

Noting that $x_{t+1} = x_t + \eta_{t+1}$ and $y_{t+1} = x_t + \eta_{t+1} + \varepsilon_{t+1}$, we see that:

$$E_{t+1}[y_{t+j+1}] - E_t[y_{t+j+1}] = \begin{cases} x_t + \eta_{t+1} + \varepsilon_{t+1} - x_t & j = 0 \\ x_t + \eta_{t+1} - x_t & j \geq 1 \end{cases}$$

Which obviously simplifies to:

$$E_{t+1}[y_{t+j+1}] - E_t[y_{t+j+1}] = \begin{cases} \eta_{t+1} + \varepsilon_{t+1} & j = 0 \\ \eta_{t+1} & j \geq 1 \end{cases}$$

With that in mind, we can conveniently express $\Delta c_{t+1} \equiv c_{t+1} - c_t$ as:

$$\begin{aligned}
\Delta c_{t+1} &= (1-\beta) \sum_{j=0}^{\infty} \beta^j \left(E_{t+1}[y_{t+j+1}] - E_t[y_{t+j+1}]\right) \\
&= (1-\beta)\beta^0 \left(E_{t+1}[y_{t+0+1}] - E_t[y_{t+0+1}]\right) + (1-\beta) \sum_{j=1}^{\infty} \beta^j \left(E_{t+1}[y_{t+j+1}] - E_t[y_{t+j+1}]\right) \\
&= (1-\beta)\left(\eta_{t+1} + \varepsilon_{t+1}\right) + (1-\beta) \sum_{j=1}^{\infty} \beta^j \eta_{t+1} \\
&= (1-\beta)\left(\varepsilon_{t+1}\right) + (1-\beta)\eta_{t+1} + (1-\beta) \sum_{j=1}^{\infty} \beta^j \eta_{t+1} \\
&= (1-\beta)\left(\varepsilon_{t+1}\right) + (1-\beta) \sum_{j=0}^{\infty} \beta^j \eta_{t+1} \\
&= (1-\beta)\left(\varepsilon_{t+1}\right) + \eta_{t+1}(1-\beta) \sum_{j=0}^{\infty} \beta^j \\
&= (1-\beta)\left(\varepsilon_{t+1}\right) + \eta_{t+1}(1-\beta) \frac{1}{1-\beta} \\
&= (1-\beta)\varepsilon_{t+1} + \eta_{t+1}
\end{aligned}$$

If we define $u_t = (1-\beta)\varepsilon_{t+1} + \eta_{t+1} \sim \mathcal{N}(0, (1-\beta)^2\sigma_\varepsilon^2 + \sigma_\eta^2)$, then we can think of $\Delta c_t$ as white noise process with mean 0 and variance: $(1-\beta)^2\sigma_\varepsilon^2 + \sigma_\eta^2)$.

From this, it is clear to see that the IRFs and autocovariances for all $j \neq 0$ will be zero. There is absolutely no information passed between periods (at least for the per-period change in consumption).

If $j = 0$, then we have already expressed the variance as $(1-\beta)^2\sigma_\varepsilon^2 + \sigma_\eta^2)$ Clearly, these are increasing in the variances of the primitives. But interesting as the agent becomes more patient ($\beta \to 1$), the variance in consumption reduces to only variance in the permanent component (because people no longer respond to the transitory shocks.)

And the IRFs are simply the partials with respect to $\varepsilon$ which is $(1-\beta)$ and for $\eta$ which is 1.

Assuming $\beta \in (0,1)$ (people strictly discount the future), then the magnitude of a transitory shock on the change in consumption is less than the same sized shock to permanent income. The permanent IRFs are invariant to model parameters, whereas as individuals become more patient $\beta \to 1$, the IRF for the transitory shock approaches zero (little change to previous period consumption).

## 2. The Vector Process

Noting that

$$\begin{aligned}\Delta y_{t+1} &= (x_{t+1} + \varepsilon_{t+1}) - (x_t + \varepsilon_t) \\ &= (x_t + \eta_{t+1} + \varepsilon_{t+1}) - (x_t + \varepsilon_t) \\ &= \eta_{t+1} + \varepsilon_{t+1} - \varepsilon_t\end{aligned}$$

Decrementing the indices by one period, we can see that:

$$\begin{bmatrix}\Delta c_t \\ \Delta y_t\end{bmatrix} = \begin{bmatrix}(1-\beta)\varepsilon_t + \eta_t \\ \eta_t + \varepsilon_t - \varepsilon_{t-1}\end{bmatrix}$$

From this form, it would be natural to represent using matrix notation:

$$\mathbf{\Delta}_t = \mathbf{\Psi}_0\mathbf{u}_t + \mathbf{\Psi}_1\mathbf{u}_{t-1}$$

Where

$$\mathbf{\Psi}_0 = \begin{bmatrix}(1-\beta) & 1 \\ 1 & 1\end{bmatrix}, \mathbf{\Psi}_1 = \begin{bmatrix}0 & 0 \\ -1 & 0\end{bmatrix}, \mathbf{u}_t \sim WN\left(\mathbf{0}, \begin{bmatrix}\sigma_\varepsilon^2 & 0 \\ 0 & \sigma_\eta^2\end{bmatrix}\right)$$

Which is a rather simple vector moving average equation, but it still suffers from an identification problem because it has 8 parameters including some on the contemporaneous shocks. One way to take care that is to redefine the contemporaneous shocks so that they're no longer orthogonal but there is no longer a $\mathbf{\Psi}_0$ matrix.

Defining $\mathbf{e}_t \equiv \mathbf{\Psi}_0\mathbf{u}_t$, linearity of the normal family shows that:

$$\begin{aligned}\mathbf{e}_t &\sim WN\left(\mathbf{0}, \begin{bmatrix}(1-\beta) & 1 \\ 1 & 1\end{bmatrix}\begin{bmatrix}\sigma_\varepsilon^2 & 0 \\ 0 & \sigma_\eta^2\end{bmatrix}\begin{bmatrix}(1-\beta) & 1 \\ 1 & 1\end{bmatrix}'\right) \\ &\sim WN\left(\mathbf{0}, \begin{bmatrix}(1-\beta)\sigma_\varepsilon^2 & \sigma_\eta^2 \\ \sigma_\varepsilon^2 & \sigma_\eta^2\end{bmatrix}\begin{bmatrix}(1-\beta) & 1 \\ 1 & 1\end{bmatrix}\right) \\ &\sim WN\left(\mathbf{0}, \begin{bmatrix}(1-\beta)^2\sigma_\varepsilon^2 + \sigma_\eta^2 & (1-\beta)\sigma_\varepsilon^2 + \sigma_\eta^2 \\ (1-\beta)\sigma_\varepsilon^2 + \sigma_\eta^2 & \sigma_\varepsilon^2 + \sigma_\eta^2\end{bmatrix}\right)\end{aligned}$$

and that we solve for last period's shocks:

$$\mathbf{e}_{t-1} = \mathbf{\Psi}_0\mathbf{u}_{t-1} \implies \mathbf{u}_{t-1} = \mathbf{\Psi}_0^{-1}\mathbf{e}_{t-1}$$

With that we can re-express our process as:

$$\mathbf{\Delta}_t = \mathbf{e}_t + \mathbf{\Psi}_1\mathbf{\Psi}_0^{-1}\mathbf{e}_{t-1}$$

Noting that using the familiar $2 \times 2$ inverse identity, we see that:

$$\begin{aligned}
\mathbf{\Psi}_0^{-1} &= \frac{1}{\det(\mathbf{\Psi}_0)} \begin{bmatrix} 1 & -1 \\ -1 & (1-\beta) \end{bmatrix} \\
&= \frac{1}{(1-\beta)\cdot 1 - 1 \cdot 1} \begin{bmatrix} 1 & -1 \\ -1 & (1-\beta) \end{bmatrix} \\
&= \frac{1}{-\beta} \begin{bmatrix} 1 & -1 \\ -1 & (1-\beta) \end{bmatrix}
\end{aligned}$$

And combining $\mathbf{\Psi}_1 \mathbf{\Psi}_0^{-1}$ into a single matrix $\tilde{\mathbf{\Psi}}_1$, we see that:

$$\begin{aligned}
\tilde{\mathbf{\Psi}}_1 &= \mathbf{\Psi}_1 \mathbf{\Psi}_0^{-1} \\
&= \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} \frac{1}{-\beta} \begin{bmatrix} 1 & -1 \\ -1 & (1-\beta) \end{bmatrix} \\
&= \frac{1}{-\beta} \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & (1-\beta) \end{bmatrix} \\
&= \frac{1}{-\beta} \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 \\ \frac{1}{\beta} & \frac{1}{-\beta} \end{bmatrix}
\end{aligned}$$

So the final form for the vector moving average process is:

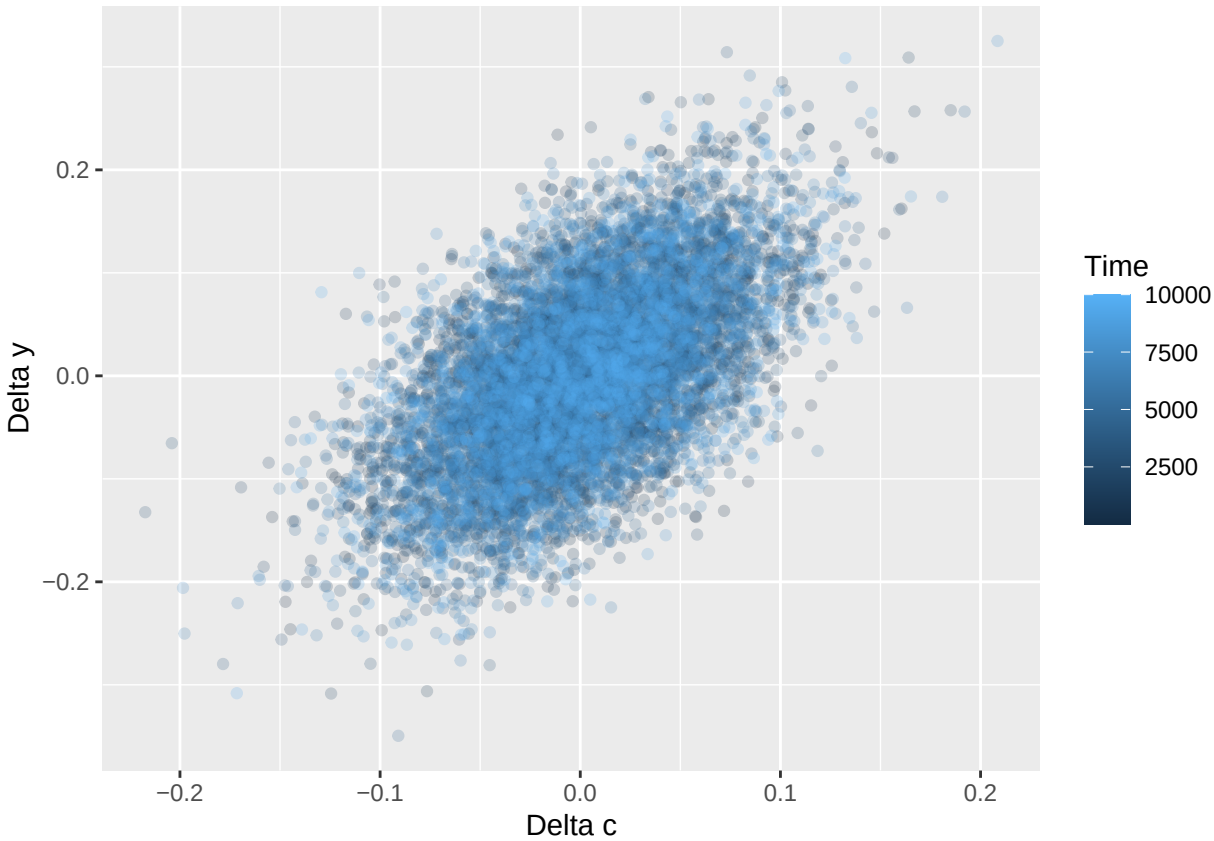
$$\mathbf{\Delta}_t = \mathbf{e}_t + \tilde{\mathbf{\Psi}}_1 \mathbf{e}_{t-1}$$
$$s.t.$$
$$\tilde{\mathbf{\Psi}}_1 = \begin{bmatrix} 0 & 0 \\ \frac{1}{\beta} & \frac{1}{-\beta} \end{bmatrix}, \mathbf{e}_t \sim WN\left(\mathbf{0}, \begin{bmatrix} (1-\beta)^2\sigma_\varepsilon^2 + \sigma_\eta^2 & (1-\beta)\sigma_\varepsilon^2 + \sigma_\eta^2 \\ (1-\beta)\sigma_\varepsilon^2 + \sigma_\eta^2 & \sigma_\varepsilon^2 + \sigma_\eta^2 \end{bmatrix}\right)$$

## 3. Simulation Exercise

The code and results below show that even after 10,000 simulations the model fails to converge to parameters given above. This is likely because the $\tilde{\mathbf{\Psi}}_1$ matrix is singular which makes it hard to identify all of the relevant parameters.

```
capT = 10e3
beta = .99
sigma_e = .05
sigma_h = .05

e_draws <- rnorm(capT)*sigma_e
h_draws <- rnorm(capT)*sigma_h

rand_mat <- cbind(e = e_draws, h = h_draws)
v_mat <- matrix(nrow = nrow(rand_mat), ncol = 2)
colnames(v_mat) <- c("Delta c","Delta y")

for (i in 1:capT){
  if (i == 1){
    rand_last <- c(e = 0, h = 0)
  }else{
    rand_last <- rand_mat[i-1,]
```

```r
  }

  rand_cur <- rand_mat[i,]

  v_mat[i, 'Delta c'] <- (1-beta)*rand_cur['e']+rand_cur['h']
  v_mat[i, 'Delta y'] <- rand_cur['h'] + rand_cur['e'] - rand_last['e']
}

library(ggplot2)
ggplot()+
  geom_point(aes(x=v_mat[,1], y = v_mat[,2], col = 1:capT), alpha = .2)+
  labs(color = "Time",
       x = "Delta c",
       y = "Delta y")
```




```r
library(MTS)
m2=VARMA(v_mat,p=0,q=1,include.mean=FALSE)
```

```
## Number of parameters:  4
## initial estimates:  0.0142 -0.0119 0.9312 -0.9245
## Par. lower-bounds:  -0.0136 -0.0308 0.8905 -0.9522
## Par. upper-bounds:  0.042 0.007 0.9719 -0.8968
## Final   Estimates:  -0.006954609 0.007020808 0.954407 -0.9521864
##
## Coefficient(s):
##       Estimate  Std. Error  t value Pr(>|t|)
## [1,] -0.006955    0.014123   -0.492    0.622
```

```
## [2,]  0.007021    0.009865    0.712    0.477
## [3,]  0.954407    0.026800   35.613   <2e-16 ***
## [4,] -0.952186    0.024020  -39.641   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## ---
## Estimates in matrix form:
## MA coefficient matrix
## MA( 1 )-matrix
##          [,1]     [,2]
## [1,]  0.00695 -0.00702
## [2,] -0.95441  0.95219
##
## Residuals cov-matrix:
##             [,1]        [,2]
## [1,] 0.002386426 0.002432595
## [2,] 0.002432595 0.004976446
## ----
## aic=  -12.02991
## bic=  -12.02702
```

```r
coef_target <- matrix(c(0, 0, 1/beta, -1/beta), nrow=2, byrow = T)


sigma_target <- matrix(c(
  (((1-beta)^2)*(sigma_e^2))+(sigma_h^2), ((1-beta)*(sigma_e^2))+(sigma_h^2),
  ((1-beta)*(sigma_e^2))+(sigma_h^2), (sigma_e^2)+(sigma_h^2)
  ), byrow = T, nrow = 2)
```

```
## Pct Error in Coefs
```

```
##          [,1]        [,2]
## [1,]  -100.00 -100.000000
## [2,] 14287.25    6.082272
```

```
## Pct Error in Sigma Matrix
```

```
##          [,1]      [,2]
## [1,] 4.769662 3.7986227
## [2,] 3.798623 0.4733049
```

# The Kalman Filter

## 1. Analytical Moments

We are given the ARMA(1,1) process:

$$y_t = \phi y_{t-1} + \varepsilon_t + \gamma \varepsilon_{t-1}$$

In lag operator notation, we have:

$$\begin{aligned} y_t &= \phi L y_t + \varepsilon_t + \gamma L \varepsilon_t \\ (1 - \phi L) y_t &= (1 + \gamma L) \varepsilon_t \end{aligned}$$

The multiplicative inverse of $(1-\phi L)$ is $(1+\phi L+\phi^2 L^2-\ldots)$, and so we can alternately express the ARMA process as an MA($\infty$):

$$\begin{aligned}
(1+\phi L+\phi^2 L^2-\ldots)(1-\phi L)y_t &= (1+\phi L+\phi^2 L^2-\ldots)(1+\gamma L)\varepsilon_t \\
y_t &= (1+\phi L+\phi^2 L^2-\ldots)(1+\gamma L)\varepsilon_t \\
&= [(1+\phi L+\phi^2 L^2-\ldots)+(\gamma L+\gamma\phi L+\gamma\phi^2 L^2+\ldots)]\varepsilon_t \\
&= [\underbrace{1}_{\psi_0}+\underbrace{\phi^0(\phi+\gamma)}_{\psi_1}L-\underbrace{\phi^1(\phi+\gamma)}_{\psi_2}L^2-\underbrace{\phi^2(\phi+\gamma)}_{\psi_3}L^3+\ldots]\varepsilon_t \\
&= \psi(L)\varepsilon_t
\end{aligned}$$

Assuming at every point that each step is well defined.

In this form, it is much easier to compute the various moments of the distribution.

Namely we have:

$$\begin{aligned}
E[y_t] &= E\left[[1+\phi^0(\phi+\gamma)L+\phi^1(\phi+\gamma)L^2+\phi^2(\phi+\gamma)L^3-\ldots]\varepsilon_t\right] \\
&= [1+\phi^0(\phi+\gamma)L+\phi^1(\phi+\gamma)L^2+\phi^2(\phi+\gamma)L^3-\ldots]E[\varepsilon_t] \\
&= [1+\phi^0(\phi+\gamma)L+\phi^1(\phi+\gamma)L^2+\phi^2(\phi+\gamma)L^3-\ldots]0 \\
&= 0
\end{aligned}$$

The variance of any $MA(\infty)$ process is given by:

$$\begin{aligned}
\gamma_0 &= \sigma^2\sum_{j=0}^{\infty}\psi_j^2 \\
&= \sigma^2\left(\psi_0^2+\sum_{j=1}^{\infty}\psi_j^2\right) \\
&= \sigma^2\left(1^2+\sum_{j=1}^{\infty}(\phi^{j-1}(\phi+\gamma))^2\right) \\
&= \sigma^2\left(1+(\phi+\gamma))^2\sum_{j=1}^{\infty}\phi^{j-1\,2}\right) \\
&= \sigma^2\left(1+(\phi+\gamma))^2\sum_{j=0}^{\infty}\phi^{j\,2}\right) \\
&= \sigma^2\left(1+(\phi+\gamma))^2\sum_{j=0}^{\infty}\phi^{2j}\right) \\
&= \boxed{\sigma^2\left(1+(\phi+\gamma))^2\frac{1}{1-\phi^2}\right)}
\end{aligned}$$

requiring that $\phi^2<1 \implies |\phi|<1$

The $j$th autocovariance of the MA($\infty$) is given by:

$$
\begin{aligned}
\gamma_j &= \sigma^2 \sum_{k=0}^{\infty} \psi_k \psi_{j+k} \\
&= \sigma^2 \left( \psi_0 \psi_j + \sum_{k=1}^{\infty} \psi_k \psi_{j+k} \right) \\
&= \sigma^2 \left( 1 \cdot \phi^{j-1}(\phi+\gamma) + \sum_{k=1}^{\infty} \phi^{k-1}(\phi+\gamma)\phi^{j+k-1}(\phi+\gamma) \right) \\
&= \sigma^2 \left( 1 \cdot \phi^{j-1}(\phi+\gamma) + \sum_{\kappa=0}^{\infty} \phi^{\kappa}(\phi+\gamma)\phi^{j+\kappa}(\phi+\gamma) \right) \\
&= \sigma^2 \left( 1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \sum_{\kappa=0}^{\infty} \phi^{\kappa}\phi^{j+\kappa} \right) \\
&= \sigma^2 \left( 1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \sum_{\kappa=0}^{\infty} \phi^{2\kappa}\phi^{j} \right) \\
&= \sigma^2 \left( 1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \phi^{j} \sum_{\kappa=0}^{\infty} \phi^{2\kappa} \right) \\
&= \sigma^2 \left( 1 \cdot \phi^{j-1}(\phi+\gamma) + (\phi+\gamma)^2 \phi^{j} \frac{1}{1-\phi^2} \right) \\
&= \sigma^2 \phi^{j-1}(\phi+\gamma) \left( 1 + (\phi+\gamma)\phi \frac{1}{1-\phi^2} \right) \\
&= \sigma^2 \phi^{j-1}(\phi+\gamma) \left( 1 + (\phi+\gamma) \frac{\phi}{1-\phi^2} \right) \\
&= \sigma^2 \phi^{j-1}(\phi+\gamma) \left( \frac{1-\phi^2}{1-\phi^2} + \frac{\phi^2+\phi\gamma}{1-\phi^2} \right) \\
&= \boxed{\sigma^2 \phi^{j-1}(\phi+\gamma) \left( \frac{1+\phi\gamma}{1-\phi^2} \right)}
\end{aligned}
$$

Truly a bizarre expression that leaves little room for intuition. However, the series convergence there also only required that $\phi^2 < 1$, so $|\phi| < 1$ is still sufficient to ensure covariance stationarity.

## 2. Conditional Expectations

## 3. Conditional Likelihood

## 4. State Space Representation

A state space representation is a system of vector equations of the form:

$$
\begin{aligned}
y_t &= \mu + A z_t + H \eta_t \\
z_t &= \Phi z_{t-1} + R \xi_t
\end{aligned}
$$

Where $\eta_t \sim \mathcal{N}(0, I), \xi_t \sim \mathcal{N}(0, \Sigma)$. Letting $z_t = \begin{bmatrix} y_t & \varepsilon_t \end{bmatrix}'$ we can represent the observation equation as:

$$
y_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_t \\ \varepsilon_t \end{bmatrix} + 0\eta_t
$$

The $\eta$ component will not be needed, so we can set $H = 0$ . Here's a canonical way to represent the state equation that ensures dimensional conformability at every step.

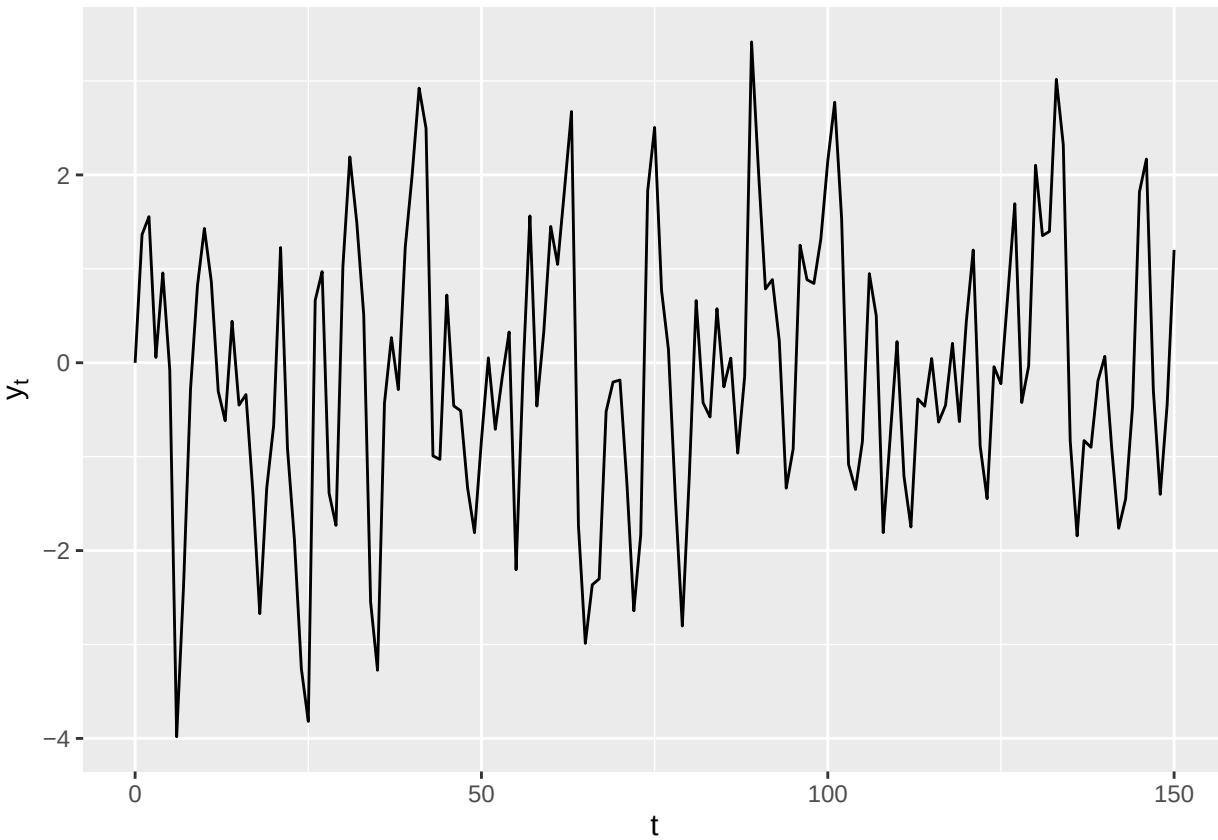
$$\begin{bmatrix} y_t \\ \varepsilon_t \end{bmatrix} = \begin{bmatrix} \phi & \gamma \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ \varepsilon_{t-1} \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \xi_t$$

Where $\xi_t$ is a 2-long vector distributed $\xi_t \sim \mathcal{N}(0, \sigma^2 I)$. You could also use a one dimensional $\xi_t$ and then just the first column of the matrix before $\xi_t$.

## 5. Simulation

```
capT = 150
# structural parameters
phi = .9
gamma = .1
sigma_squared = 1

# state space parameters
dim_state <- 2
Sigma = sigma_squared * diag(nrow = dim_state)
R = matrix(c(1, 0,
             1, 0),
           byrow = T, nrow = dim_state)
Phi = matrix(c(gamma, phi,
               0, 0),
             byrow = T, nrow = dim_state)

# Observation parameters
dim_obs <- 1 #
mu <- c(0)
A <- matrix(c(1, 0), byrow = T, nrow = dim_obs, ncol = dim_state)
H <- matrix(c(0), byrow = T, nrow = dim_obs)

evolve_state <- function(z, Phi, R, Sigma){
  # compute multivariate normal using linearity
  xi = Sigma%*%rnorm(nrow(Sigma))

  (Phi %*% z) + R%*%xi
}

evolve_observations <- function(z, mu, A, H){
  eta <- rnorm(nrow(H))

  mu + (A %*% z) + H%*%eta
}

z0 = c(0,0)
simulate <- function(capT, z0, Phi, R, Sigma, mu, A, H){
  # loop capT times, feeding the result of the last iteration into
  # this this iteration
  states <- purrr::accumulate(
  1:capT,
  ~ evolve_state(.x, Phi, R, Sigma),
  .init = z0
)
```

```r
  # compute observation from latent state
  observations <- lapply(states, function(z){
    evolve_observations(z, mu, A, H)
  })

  # return long matrices with one row per time step
  states <- do.call(cbind, states) |> t()
  observations = do.call(cbind, observations) |> t()


  return(list(states = states, observations = observations))
}

sim_results <- simulate(capT, c(0,0), Phi, R, Sigma, mu, A, H)

library(ggplot2)
ggplot()+
  geom_line(aes(x=0:(nrow(sim_results$observations)-1),
                y=sim_results$observations[,1]))+
  labs(y = expression("y"[t]*""),
       x = "t")
```




## 6. Implementing the Kalman Filter

```r
predict_y <- function(z, mu, A){
  mu + (A %*% z)
```

```
}

compute_v <- function(y, z, mu, A){
 y - predict_y(z, mu, A)
}

v_posterior_variance <- function(P, A, H){
  (A %*% P %*% t(A)) + H%*%t(H)
}

prediction_error_variance <- function(z, P, A){
  E
}
```

### 7. Applying the Kalman Filter

## Metropolis Hastings

### 1. Numerical Maximization

### 2. Generating M-H Draws

### 3. Lots of Simulations